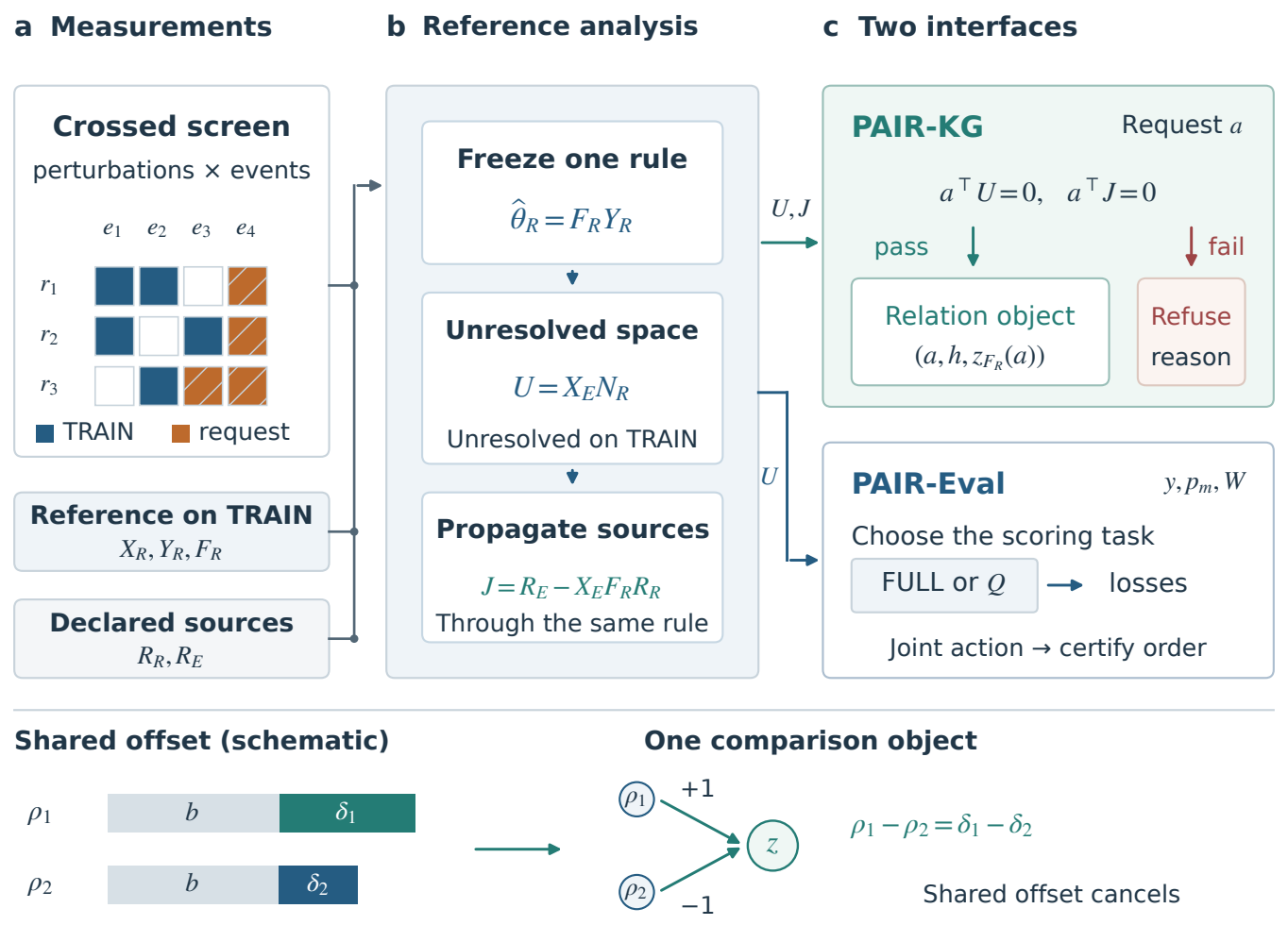
a Measurements
b Reference analysis
c Two interfaces
Crossed screen
perturbations × events
e_1 e_2 e_3 e_4
r_1
r_2
r_3
TRAIN
request
Reference on TRAIN
X_R, Y_R, F_R
Declared sources
R_R, R_E
Freeze one rule
$\hat{\theta}_R = F_R Y_R$
Unresolved space
$U = X_E N_R$
Unresolved on TRAIN
Propagate sources
$J = R_E - X_E F_R R_R$
Through the same rule
U, J
U
PAIR-KG
Request a
$a^\top U = 0, \quad a^\top J = 0$
pass
fail
Relation object
$(a, h, z_{F_R}(a))$
Refuse
reason
PAIR-Eval
y, p_m, W
Choose the scoring task
FULL or Q
losses
Joint action → certify order
Shared offset (schematic)
ρ_1
b
δ_1
ρ_2
b
δ_2
One comparison object
ρ_1
ρ_2
+1
−1
z
$\rho_1 - \rho_2 = \delta_1 - \delta_2$
Shared offset cancels